



Liquidity, Activism Disclosure, and Takeover Premia

Exit, Voice, and Corporate Control

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Three empirical facts motivate the paper.

- Brav et al., 2008: activist campaigns often culminate in corporate control events.
- Greenwood and Schor, 2009: a large share of activist returns comes through takeover outcomes.
- Back et al., 2018: pre-filing trading suggests activists exploit informational advantages before disclosure.

Gap

No existing framework combines **endogenous exit vs. voice choice, order-flow inference**, and **takeover entry from public market signals**.

Question

How does market liquidity shape *takeover premia* for minority shareholders when a blockholder can *exit, hold, or engage*?

Closest inspiration

Edmans et al., 2015: public market information feeds back into takeover outcomes. This paper adds endogenous exit/voice choice and disclosure.

Bridge: exit / voice, microstructure, and corporate control.

Institutional trigger

- US: Schedule 13D at 5% beneficial ownership
- UK: 3% threshold with tighter timing
- Crossing threshold makes activism *publicly observable*

Quiet Voice

Engage below threshold

No filing signal

Inference from X

Public Voice

Stake building + engage

Filing reveals activism

Baseline: $D = 1$ reveals public intervention

What the market learns

- Before activism is fully public, trading and disclosure leave a public footprint.
- Without a filing, order flow makes activism look more or less likely.
- Liquidity matters because it changes how informative that footprint is.

Why takeover outcomes move

- When activism looks more likely, expected standalone value is higher.
- Expected takeover premia are also higher.
- But some bidders stay out because the target looks harder or more expensive to acquire.
- Minority takeover gains reflect the balance of those forces.

In short: the same public footprint that raises expected value and premia can also discourage bidder entry, and takeover gains reflect that balance.

Mechanism logic ↗

Full posteriors ↗

Bid deterrence ↗

Choices (blockholder)

s — private signal

q — trade order $\in \{-1, 0, +1\}$

a — engage $\in \{0, 1\}$

$C(s)$ — engagement cost

k_1 — Exit/Hold cutoff

k_0 — Hold/Quiet Voice cutoff

k_D — Quiet/Public Voice cutoff

ω_i — region probabilities

Outcomes (equilibrium)

$\tilde{\Delta}$ — expected value gain from engagement

$\tilde{m}$ — expected premium under engagement

$\pi(X, D)$ — posterior on $a=1$

$\hat{V}(X, D)$ — standalone value

$\tilde{m}(X, D)$ — expected premium

$p(X, D)$ — bid probability

Observables (market)

z — noise trade $\in \{-1, 0, +1\}$

κ — liquidity (noise intensity)

X — order flow $= q + z$

D — disclosure $= \mathbf{1}\{q=+1\}$

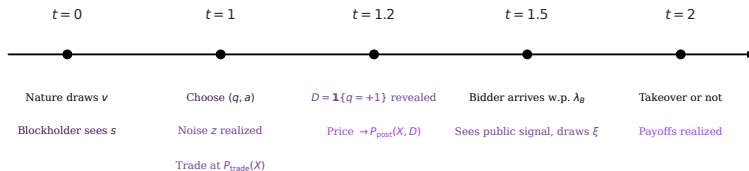
ξ — bidder synergy shock

s_ξ — synergy dispersion

P_{post} — post-disclosure price

P_{trade} — execution price

Δ^{min} — minority takeover gains



Action menu

Action	(q, a)
Exit	$(-1, 0)$
Hold	$(0, 0)$
Quiet Voice	$(0, 1)$
Public Voice	$(+1, 1)$

Noise trading (liquidity κ)

$z \in \{-1, 0, 1\}$, $\mathbb{P}(z=0) = 1 - \frac{2}{3}\kappa$, $\mathbb{P}(z=\pm 1) = \frac{\kappa}{3}$.

Bidder conditions on the public market signal; in the baseline, it is summarized by (X, D) .

Identification in the baseline

- Public Voice is the only threshold-crossing action:
 $(q, a) = (+1, 1) \Rightarrow D = 1$.
- Hold and Quiet Voice both imply $q = 0$, so $D = 0$ states are separated only by inference.

What outsiders see

- Blockholder strategy is a monotone cutoff rule $\sigma_B(s)$, summarized by $k_1 \leq k_0 \leq k_D$.
- Outsiders do not observe the blockholder's signal s or exact motive.
- They see only the public market signal left by trading and disclosure; in the baseline, that signal is summarized by (X, D) .
- From that public signal, they infer which actions are more likely.

Full PBE ↗

Bayes ↗

What equilibrium requires

- The market maker sets prices from those inferences.
- The bidder decides whether to enter using the same public signal and private synergy ξ .
- The cutoff rule must be optimal given those pricing and bidding responses.
- Beliefs, prices, bids, and cutoffs must fit together in a self-consistent way.

Public state ↗

Existence ↗

Action structure

- A1: Interior regions; no single action always dominates.
- A2: Engagement cost $C(s)$ is positive and decreasing in s .
- A3: Engagement raises the expected premium, $\tilde{m} > m_0$.

Bidder entry

- A4: Synergy bounds keep bid incidence interior.
- A5: $\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$ (net deterrence).

Full assumptions ↗

Regularity

- A6: The cutoff map selects a unique fixed point numerically.
- A7: Bidder arrival is bounded, $\lambda_B \leq 1/2$, supporting single-crossing.

Why these matter

- A1–A3 generate the four action regions.
- A4–A5 keep bids interior and make inferred engagement deter bids.
- A6–A7 discipline equilibrium selection.

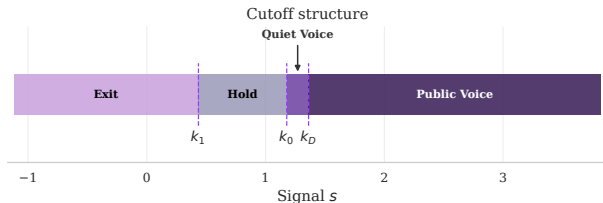
A5 and bid deterrence ↗

Proposition 1. Equilibrium behavior is organized by ordered cutoffs $k_1 \leq k_0 \leq k_D$.

Signal Region	Action
$s < k_1$	Exit
$k_1 \leq s < k_0$	Hold
$k_0 \leq s < k_D$	Quiet Voice
$s \geq k_D$	Public Voice

Action map: Exit = $(-1, 0)$, Hold = $(0, 0)$,
Quiet Voice = $(0, 1)$, Public Voice = $(+1, 1)$.

Disclosure: $D = 1$ iff $q = +1$.
Hold and Quiet Voice both imply $q = 0$, so they share the same order-flow distribution.



[Proof sketch ↗](#)

[Action set ↗](#)

[Cutoffs vs. \$\kappa\$ ↗](#)

Inference

- Disclosed branch: $D = 1 \Rightarrow \pi(X, 1) = 1$.
- Nondisclosed branch: $D = 0$ mixes Exit, Hold, and Quiet Voice through order flow X .
- Liquidity κ matters by changing how informative X is in nondisclosed states.
- Channel: X changes π , which moves $\bar{m}$, p , and $\Delta^{\min}$.

Discrete flow ↗

Full posteriors ↗

Bidder and prices

$$\hat{V} = \mathbb{E}[v \mid X, D] + \tilde{\Delta} \pi, \quad \bar{m} = m_0 + (\tilde{m} - m_0)\pi$$

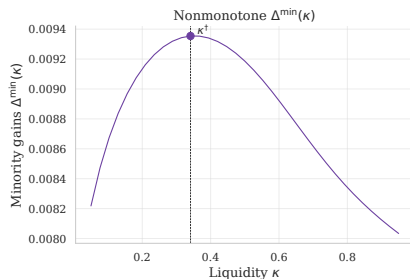
$$p = \lambda_B(1 - \Lambda(\Gamma)), \quad \Gamma = \frac{\hat{V} + \bar{m} + K - (\bar{S} + \pi \Delta_S)}{s_\xi}$$

$$P_{\text{post}} = \delta(\hat{V} + p \bar{m}), \quad P_{\text{trade}}(X) = \mathbb{E}[P_{\text{post}} \mid X]$$

- Under A5 (net deterrence), higher inferred engagement raises value and premia enough to deter marginal bids.
- Sequential mechanism: public signal $\rightarrow$ beliefs $\rightarrow$ bidder entry $\rightarrow$ prices.

Mechanism logic ↗

Bid deterrence ↗



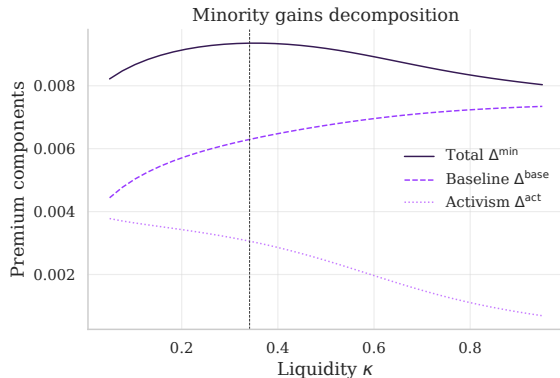
Baseline result. Minority takeover gains $\Delta^{\min}(\kappa) = \mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}]$ are hump-shaped, with an interior maximizer.

Interpretation

- More liquidity improves camouflage and lowers adverse selection.
- More liquidity weakens inference and erodes the activism premium.
- The hump is a quantitative result, not a closed-form theorem.

Endpoints ↗

Proof idea ↗



Exact decomposition

$$\Delta^{\text{base}}(\kappa) = m_0 \mathbb{P}(\text{bid}),$$

$$\Delta^{\text{act}}(\kappa) = (\tilde{m} - m_0) \mathbb{E}[\pi \cdot \mathbf{1}\{\text{bid}\}],$$

$$\Delta^{\min}(\kappa) = \Delta^{\text{base}} + \Delta^{\text{act}},$$

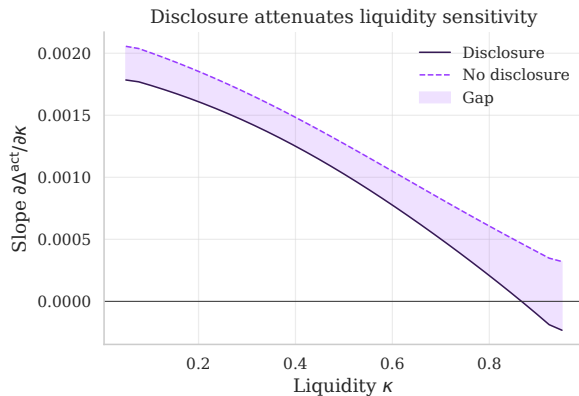
where $\pi \equiv \pi(X, D)$.

Reading the forces

- Δ^{base} rises with bid incidence.
- Δ^{act} declines as inference compresses.

Bidder sens. ↗

Activism sens. ↗



Partial-equilibrium statement

Hold **cutoffs** fixed and vary κ only through inference. Only the nondisclosed component depends on κ because $D = 1$ implies $\pi = 1$.

Main implication

Shifting probability mass toward disclosed engagement attenuates $|\partial \Delta^{\text{act}} / \partial \kappa|$.

Interpretation

- UK-style stricter thresholds reduce liquidity sensitivity.
- A no-disclosure counterfactual amplifies it.

Other regimes ↗

Noisy rumors ↗

Anonymity ↗

Takeaways

- Liquidity matters because it changes inference from order flow.
- Threshold disclosure shifts activism from inferred states to observed states.
- Minority takeover gains can peak at intermediate liquidity.

Theory status

Analytical: cutoffs, posteriors, exact decomposition, partial-equilibrium attenuation.

Numerical: baseline hump, fixed-point selection, broader disclosure and welfare patterns.

Testable implications

- Unconditional hump test in minority takeover gains.
- Decomposition: bid incidence rises with liquidity while activism-related premia weaken.
- Stricter disclosure or more observable firms should flatten the liquidity–premia relation; 13D filing CARs should be larger for more liquid targets.

Welfare caveat

The hump is about $\Delta^{\min}$, not total surplus. Welfare and general-equilibrium disclosure trade-offs remain preliminary.

Proof Sketch for Proposition 1

Key ingredients:

- Payoffs are affine in $\hat{v}(s) = \mathbb{E}[v \mid s]$; $\hat{v}$ increases in s
- Engagement cost $C(s)$ is (weakly) decreasing in s
- $P_{\text{trade}}(X)$ is independent of s within a given action

These imply strict single-crossing of adjacent payoff differences, yielding a monotone best response with cutoffs $k_1 \leq k_0 \leq k_D$.

Baseline action-set interpretation

The model is written for a control-oriented activist, so the only threshold-crossing action in the baseline action set is Public Voice. Appendix lemma: passive threshold crossing is dominated for sufficiently high signals, and in the baseline calibration $k_D \geq \bar{s}$.

← Slide 8: Cutoffs

Existence (Proposition 4)

Existence is stated under maintained regularity conditions: a compact ordered cutoff set Θ , a well-defined cutoff mapping T , and continuity on reached information sets with an off-path completion. Brouwer then implies a fixed point.

Uniqueness (Assumption A6)

No analytic uniqueness theorem is claimed. The solver converges to the same fixed point from multiple starts in all reported exercises.

Solver Implementation

Damped fixed-point iteration with `scipy.optimize.brentq` for sub-problems. Multi-start search with collapsed-hold fallback. Convergence tolerance: 10^{-6} .

← Slide 6: How the Model Closes

Equilibrium object

A monotone-cutoff Perfect Bayesian Equilibrium is a tuple

$$\left(k_1, k_0, k_D, \mu(\cdot | h), P_{\text{trade}}(X), P_{\text{post}}(h), \tilde{p}(h, \xi) \right)$$

such that the following conditions hold.

1. **Blockholder optimality:** the cutoff rule induced by (k_1, k_0, k_D) maximizes expected payoff at every signal s .
2. **Bidder best response:** for every reached public state h and private synergy ξ , entry follows the surplus-maximizing rule $\tilde{p}(h, \xi)$.
3. **Competitive pricing:** $P_{\text{trade}}(X) = \delta \mathbb{E}[Y | X]$ and $P_{\text{post}}(h) = \delta \mathbb{E}[Y | h]$.
4. **Belief consistency:** $\mu(\cdot | h)$ is Bayes-consistent on each reached public state.
5. **Cutoff indifference:** adjacent-action indifference equalities hold at k_1, k_0, k_D .

Public state. Let h denote the public information state at $t = 1.5$. In the baseline, h is indexed by (X, D) .

On-path belief formula

For any reached public state h with $\mathbb{P}(h) > 0$,

$$\mu(a \mid h) = \frac{\mathbb{P}(a, h)}{\mathbb{P}(h)}.$$

Equivalently, beliefs over Exit / Hold / Quiet Voice / Public Voice are pinned down by the cutoff strategy and the noise distribution via Bayes' rule.

Reached states in the baseline

- Disclosed branch: $D = 1$ implies Public Voice, so only $(0, 1)$, $(1, 1)$, $(2, 1)$ can occur.
- Nondisclosed branch: $D = 0$ mixes Exit, Hold, and Quiet Voice through order flow.
- Off-path states require a completion for existence, but the quantitative objects use on-path beliefs.

Reduced-form interpretation

The bidder need not literally observe raw exchange-level order flow. What the theorem package requires is that, by the entry stage, both the market maker and the bidder condition on the same *public state* h .

- In the baseline, h is indexed by the public trading/disclosure outcome (X, D) .
- This is a sufficient-statistic formulation: any richer public history generating the same posterior $\mu(\cdot \mid h)$ is observationally equivalent for prices and bidder entry.
- The economics is public learning from a trading footprint, not literal tape-reading by bidders.

← Slide 6: How the Model Closes

Let $\alpha_i = (k_i - \mu)/\sigma_s$ and inverse Mills ratios $\lambda_L(\alpha) = \phi(\alpha)/\Phi(\alpha)$, $\lambda_U(\alpha) = \phi(\alpha)/(1 - \Phi(\alpha))$.

Conditional Means (Lemma 3)

$$\mu_E = \mathbb{E}[v \mid s < k_1] = \mu - \beta\sigma_s\lambda_L(\alpha_1), \quad \mu_P = \mathbb{E}[v \mid s \geq k_D] = \mu + \beta\sigma_s\lambda_U(\alpha_D),$$

$$\mu_H = \mu + \beta\sigma_s \frac{\phi(\alpha_1) - \phi(\alpha_0)}{\Phi(\alpha_0) - \Phi(\alpha_1)}, \quad \mu_Q = \mu + \beta\sigma_s \frac{\phi(\alpha_0) - \phi(\alpha_D)}{\Phi(\alpha_D) - \Phi(\alpha_0)}.$$

These enter $\hat{V}(X, D)$ through the conditional expectation $\mathbb{E}[v \mid X, D]$, which weights $\mu_E, \mu_H, \mu_Q, \mu_P$ by the posterior action probabilities.

← Slide 9: Inference, Bids, Prices

Posterior Engagement Probabilities (Proposition 2)

Baseline disclosed branch: $D = 1 \Rightarrow \pi(X, 1) = 1$. For $D = 0$ with $p_0 = 1 - 2\kappa/3$, $p_1 = \kappa/3$:

$$\begin{aligned}\pi(1, 0) &= \frac{\omega_Q}{\omega_H + \omega_Q}, \\ \pi(-1, 0) &= \frac{\omega_Q p_1}{(\omega_H + \omega_Q)p_1 + \omega_E p_0}, \\ \pi(0, 0) &= \frac{\omega_Q p_0}{(\omega_H + \omega_Q)p_0 + \omega_E p_1}.\end{aligned}$$

Comparative statics (holding ω fixed)

$$\partial_\kappa \pi(1, 0) = 0, \quad \partial_\kappa \pi(-1, 0) \geq 0, \quad \partial_\kappa \pi(0, 0) \leq 0.$$

Step 1: Public signal to activism beliefs

Trading and disclosure generate the public signal (X, D) . The market maker and bidder update to the posterior activism belief $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$.

- $D = 1$ reveals Public Voice, so $\pi(X, 1) = 1$.
- $D = 0$ mixes Exit, Hold, and Quiet Voice through order flow X .

Step 2: Beliefs move value, premia, and bid incentives

$$\begin{aligned}\hat{V}(X, D) &= \mathbb{E}[v \mid X, D] + \tilde{\Delta} \pi(X, D), & \bar{m}(X, D) &= m_0 + (\tilde{m} - m_0) \pi(X, D). \\ p(X, D) &= \lambda_B(1 - \Lambda(\Gamma)), & \Gamma &= \frac{\hat{V} + \bar{m} + K - (\bar{S} + \pi \Delta_S)}{s_\xi}.\end{aligned}$$

Higher inferred activism raises expected value and premia; under A5 it also raises the entry cutoff Γ and deters marginal bids.

Step 3: Post-disclosure price

Conditional on the realized public signal, competitive pricing averages standalone value and expected takeover surplus:

$$P_{\text{post}}(X, D) = \delta(\hat{V}(X, D) + p(X, D)\bar{m}(X, D)).$$

Trade-date price

Before disclosure is realized, the trade-date price averages over the later disclosure branch:

$$P_{\text{trade}}(X) = \sum_{d \in \{0,1\}} \mathbb{P}(D = d \mid X) P_{\text{post}}(X, d).$$

Timing logic

- Public signal (X, D) determines beliefs π .
- Those beliefs determine bidder entry and expected premia.
- Prices then aggregate value and expected takeover surplus at each stage.

Definition

$$\Delta^{\min}(\kappa) = \mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}].$$

Decomposition

Since $m^R(a) = m_0 + (\tilde{m} - m_0) \cdot a$ and $\mathbb{E}[a \mid X, D] = \pi(X, D)$,

$$\Delta^{\min}(\kappa) = \underbrace{m_0 \mathbb{P}(\text{bid})}_{\Delta^{\text{base}}} + \underbrace{(\tilde{m} - m_0) \mathbb{E}[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}]}_{\Delta^{\text{act}}}.$$

The decomposition separates *whether* a bid occurs (extensive margin) from the *size* of the premium conditional on bidding (intensive margin).

← Slide 11: Decomposition

As $\kappa \downarrow 0$ (no noise): $\mathbb{P}(z=0) \rightarrow 1$ so $X = q$ a.s. and $P_{\text{trade}}(X) \rightarrow P_{\text{post}}(X, D(X))$.

Order flow perfectly reveals the action: $X=-1 \Rightarrow$ Exit, $X=0 \Rightarrow$ Hold/Quiet Voice, $X=+1 \Rightarrow$ Public Voice. Inference is maximally sharp; execution price collapses to post-disclosure price.

As $\kappa \uparrow 1$ (maximal noise): z becomes uniform on $\{-1, 0, 1\}$. Because the support of z is bounded:

- Extreme flows still reveal disclosure: $\mathbb{P}(D=1 \mid X=2) = 1$ and $\mathbb{P}(D=1 \mid X \in \{-2, -1\}) = 0$
- Interior flows ($X \in \{0, 1\}$) continue mixing via Bayes' rule
- Δ^{act} does **not** vanish — engagement survives in the Public Voice region

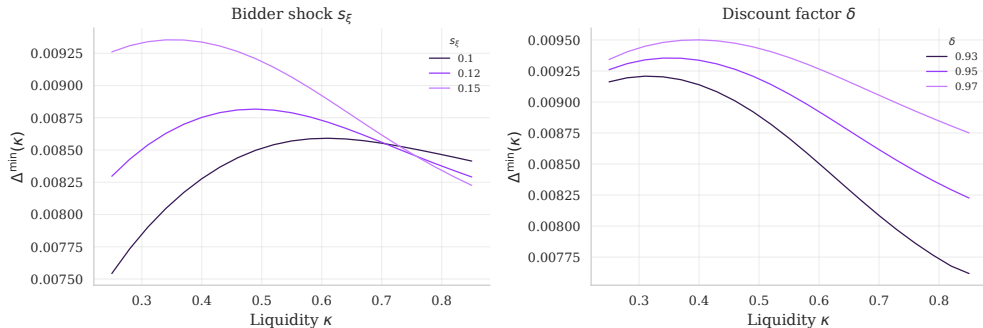
← Slide 10: Main Result

How to read Proposition 5

1. **Analytic piece:** $\Delta^{\min} = \Delta^{\text{base}} + \Delta^{\text{act}}$ is an exact decomposition.
2. **Endpoint intuition:** low and high liquidity have different information structures under bounded noise.
3. **Quantitative piece:** the interior hump is established in the baseline calibration and sensitivity exercises, not as a general closed-form theorem.

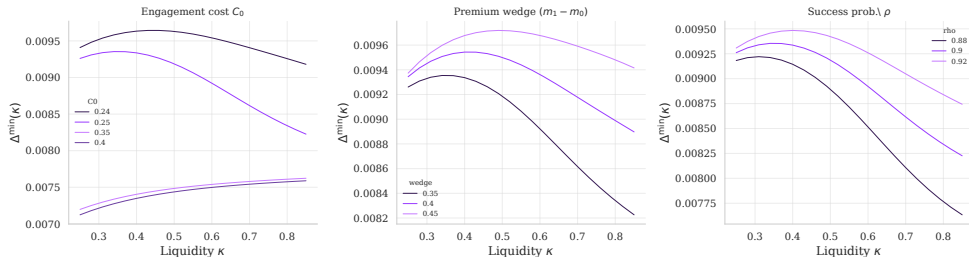
Important caveat Continuity gives an attained maximum on a compact set, but whether that maximum is interior is quantitative. The baseline calibration confirms a single-peaked hump (Figure 2); the peak can shift or disappear under other primitives.

← Slide 10: Main Result



- Larger s_{ξ} : more bidder heterogeneity flattens/shifts the hump
- Lower δ : weaker engagement incentives compress $\Delta^{\min}$

← Slide 10: Main Result

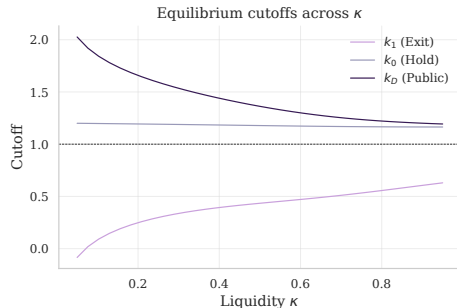


Read the panels as follows.

- Higher C_0 shrinks the voice region and can shift $\kappa^\dagger$.
- Larger $(m_1 - m_0)$ or higher ρ raise conditional premia, but can also deter bids.

← Slide 10: Main Result

B11. How do equilibrium cutoffs move with liquidity?



- Ordering $k_1 \leq k_0 \leq k_D$ holds; regions may collapse for some primitives
- Quiet Voice collapses ($k_0 = k_D$) at $\kappa \approx 0.63$ in baseline
- Baseline calibration features an interior Hold region ($k_1 < k_0$)

← Slide 8: Cutoffs

Bid probability

$$p(X, D) = \lambda_B \left(1 - \Lambda \left(\frac{\hat{V}(X, D) + \bar{m}(X, D) + K - (\bar{S} + \pi(X, D)\Delta_S)}{s_\xi} \right) \right).$$

Comparative statics

$\partial_{\bar{S}} p > 0$, $\partial_K p < 0$. Under net deterrence (A5):

$$\frac{\partial p}{\partial \pi} = -\lambda_B \Lambda'(\cdot) \frac{\tilde{\Delta} + (\tilde{m} - m_0) - \Delta_S}{s_\xi} < 0.$$

Intuition: Higher inferred engagement raises target value and the premium the bidder must pay, deterring marginal bidders.

← Slide 9: Inference, Bids, Prices

Benchmarks:

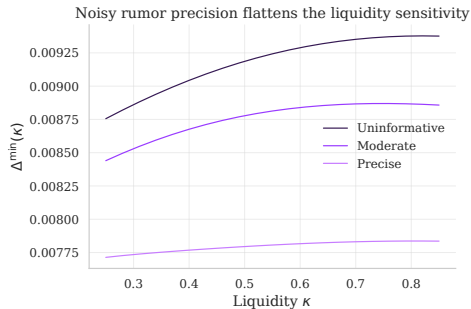
- **Full disclosure:** market observes (q, a) , inference channel shuts down
- **No disclosure:** market conditions only on X ; inference amplifies liquidity sensitivity

Posterior comparison (baseline example):

X	$\pi(X, 0)$	$\pi_{ND}(X)$	$\pi_{NR}(X, 0, R = 0)$	$\pi_{NR}(X, 0, R = 1)$	(q, a)	π_{FI}
-1	0.07	0.07	0.03	0.19	$(-1, 0)$	0.00
0	0.18	0.28	0.07	0.40	$(0, 0)$	0.00
1	0.20	0.77	0.08	0.43	$(0, 1)$	1.00
					$(1, 1)$	1.00

No disclosure raises the sensitivity of π to X ; full disclosure eliminates inference.

← Slide 12: Disclosure Attenuation



Setup Rumors add a public signal $R \in \{0, 1\}$ in nondisclosed states. The posterior becomes

$$\pi(X, 0, R) = \frac{\omega_Q \mathbb{P}(X|q=0) \mathbb{P}(R|a=1)}{\omega_E \mathbb{P}(X|q=-1) \mathbb{P}(R|a=0) + \mathbb{P}(X|q=0) (\omega_H \mathbb{P}(R|a=0) + \omega_Q \mathbb{P}(R|a=1))}$$

Takeaway More precise rumors flatten $\Delta^{\min}(\kappa)$ by reducing reliance on order-flow inference.

← Slide 12: Disclosure Attenuation

Heuristic Decomposition

$$\frac{d\Delta^{\text{act}}}{d\tau} \approx \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial \tau}}_{\text{Transparency (+)}} + \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial (\omega_Q + \omega_P)} \frac{d(\omega_Q + \omega_P)}{d\tau}}_{\text{Deterrence (-)}}.$$

Transparency Effect (+)

Shifting mass to observable states removes the inference discount. In $D = 1$ states, $\pi = 1$ and the pricing objects are κ -invariant, so the activism premium is fully capitalized.

Deterrence Effect (−)

Stricter thresholds force earlier disclosure, which can reduce expected returns to quiet engagement and shrink total activism probability $\omega_Q + \omega_P$. This is an economic heuristic, not a separate theorem.

← Slide 13: Takeaways

Comparison	
Partial Equilibrium (Prop 6)	GE heuristic
Hold (k_1, k_0, k_D) fixed	Cutoffs respond to τ
Vary κ through inference only	Vary disclosure regime τ
$ \partial \Delta^{\text{act}} / \partial \kappa \downarrow$ in ω_P	$d\Delta^{\text{act}} / d\tau$ ambiguous
Pure transparency channel	Transparency + deterrence

Key bridge: Anonymous accumulation helps keep Public Voice alive on path by letting the blockholder acquire at $P_{\text{trade}}(X)$ before the adverse-selection penalty in $P_{\text{post}}(X, 1)$ is realized.

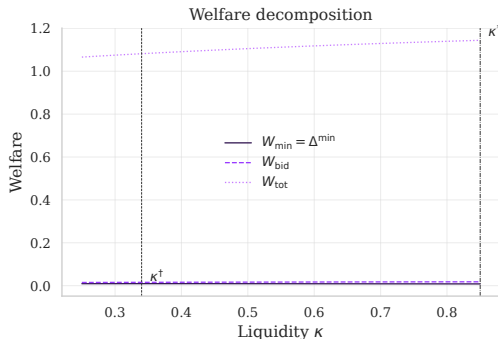
← Slide 13: Takeaways

Full reasoning chain (color-coded)

1. When $D = 1$, the action is Public Voice: $a = 1$ with certainty
2. Order flow is $X = 1 + z$, which reveals *only noise* — no information about fundamentals beyond Public Voice
3. Therefore $\pi(X, 1) = 1$ and $\mathbb{E}[v \mid X, 1] = \mu_P$ are both **constant** across $X \in \{0, 1, 2\}$
4. Since $\hat{V}$, π , and hence p are all constant, $P_{\text{post}}(X, 1)$ is invariant in X

Consequence: In the baseline action set, the only thing the market maker learns from X is the noise realization z , which is uninformative. This makes $D = 1$ a clean benchmark where inference plays no role.

← Slide 9: Inference, Bids, Prices



$\Delta^{\min}$ can peak at $\kappa^{\dagger}$ while total surplus keeps increasing. Treat this as preliminary accounting: takeover premia are *transfers* from bidder to target shareholders, so the mapping from minority gains to total surplus depends on normalization choices.

← Slide 13: Takeaways

Minority-shareholder object. With dispersed ownership, minority shareholders cannot coordinate to internalize governance benefits directly. Their main channel for capturing value from control changes is the takeover premium, consistent with the Grossman and Hart, 1980 free-rider logic.

Definition

$$\Delta^{\text{min}}(\kappa) \equiv \mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}]$$

This is the expected premium, weighted by bid occurrence, that minority shareholders receive ex ante.

Alternative metrics. Total surplus W_{tot} includes bidder surplus, so transfers cancel. It can move monotonically in κ even when Δ^{min} is hump-shaped, reflecting a different policy objective.

← Slide 10: Main Result

Parameter	Value
μ	1.00
σ_v	0.50
σ_ε	0.50
δ	0.95
κ	0.50
C_0	0.25
χ	0.50
ρ	0.90

Parameter	Value
Δ	0.25
m_0	0.10
m_1	0.45
$\bar{S}$	1.10
Δ_S	0.30
K	0.15
s_ξ	0.15
λ_B	0.20

Net deterrence (A5): $\tilde{\Delta} + (\tilde{m} - m_0) - \Delta_S = 0.225 + 0.315 - 0.30 = 0.24 > 0$. ✓

← Slide 5: Timeline

B21. Baseline equilibrium outcomes table

D	Order flow X	$P(X, D)$	$\pi(X, D)$	$p(X, D)$	$\bar{m}(X, D)$
0	-2	0.51	0.00	0.18	0.10
0	-1	0.68	0.07	0.15	0.12
0	0	0.92	0.18	0.07	0.16
0	1	0.96	0.20	0.05	0.16
1	0	1.55	1.00	0.00	0.42
1	1	1.55	1.00	0.00	0.42
1	2	1.55	1.00	0.00	0.42

$D = 1$ rows have $p = 0.00$: bids are fully deterred on the disclosed branch under A5.

← Slide 9: Inference, Bids, Prices

A1	Interior regions: no single action always dominates
A2	$C(s) > 0$ and decreasing in s
A3	Engagement raises expected premium: $\tilde{m} > m_0$
A4	Synergy bounds ensure interior bid probability: $p \in (0, 1)$
A5	Net deterrence: $\tilde{\Delta} + \tilde{m} - m_0 > \Delta_S$
A6	Cutoff map T is a contraction (verified numerically)
A7	Arrival bound $\lambda_B \leq 1/2$ (single-crossing)

Role in the model

- A1–A3 deliver four distinct action regions.
- A4–A5 keep bidder entry interior and make activism deter bids.
- A6 is the numerical uniqueness condition.
- A7 guarantees monotone payoff differences.

← Slide 7: Standing Assumptions

Sign pattern From Proposition 2 with $p_0 = 1 - 2\kappa/3$ and $p_1 = \kappa/3$:

Flow	Effect of κ	Reason
$X = 1$	none	Only Hold and Quiet Voice can generate $X = 1$ when $D = 0$, so the noise terms cancel.
$X = -1$	$\pi(-1, 0)$ rises	More noise makes sell flow more likely to be camouflage rather than Exit.
$X = 0$	$\pi(0, 0)$ falls	More noise makes zero flow more likely under Exit, lowering inferred engagement.

Interpretation. These within-regime effects generate the inference channel. Cross-equilibrium effects also move cutoffs, creating the general-equilibrium feedback in Propositions 5–7.

← Slide 9: Inference, Bids, Prices

B24. Why is passive threshold crossing omitted in the baseline action set?

Candidate omitted action: Passive Accumulation = $(+1, 0)$.

Why it is not on-path in the baseline

- Threshold crossing without engagement still triggers disclosure and the associated adverse-selection penalty.
- For high signals, the same forces that make buying attractive also make engagement attractive: expected value improvement rises while $C(s)$ falls.
- The appendix lemma shows that passive threshold crossing is dominated above a signal cutoff under maintained conditions; in the baseline calibration, the threshold-crossing region is therefore Public Voice.

Interpretation

The baseline action set is not ruling out passive buying by assumption. It is summarizing the on-path equilibrium object for a control-oriented activist: once she crosses the threshold in relevant high-signal states, she also chooses to engage.

What the discrete setup buys

- Finite order-flow states give transparent Bayes updating in both $D = 0$ and $D = 1$ regimes.
- The market-maker problem stays feed-forward once disclosure arrives after anonymous execution.
- The model can combine governance choice, disclosure, and takeover feedback without a price fixed point.

What it gives up

- No gradual multi-round accumulation.
- Bounded noise support rather than a continuous Kyle environment.
- No claim that the hump is a generic closed-form property of all trading protocols.

Interpretation: this is a tractable first-pass theory design, not a claim that real activist trading is literally one-shot.

Timing wedge

At $t = 1$: buy at $P_{\text{trade}}(X)$ $\implies$ At $t = 1.2$: $D = 1$ and price updates to $P_{\text{post}}(X, 1)$

- The activist accumulates before the market fully capitalizes observed engagement.
- If she had to buy at the fully disclosed price, she would internalize the whole activism premium immediately.
- That would shrink the Public Voice region and weaken the transparency-attenuation channel in general equilibrium.

Bridge to the disclosure results

Anonymous execution is what lets Public Voice survive on path in the baseline calibration. That is why the partial-equilibrium attenuation result has economic bite instead of being a purely mechanical accounting statement.